

SIH 26147 — Next-Gen Astonishing RF Intelligence & Demodulation Visualizations

Publication-Grade 3D & Physical Waveform DSP Showcase (300 DPI)

Advanced spectral elevation modeling, receiver eye diagrams, vector phase transitions, polar fingerprint radars, and 3D feature clustering.

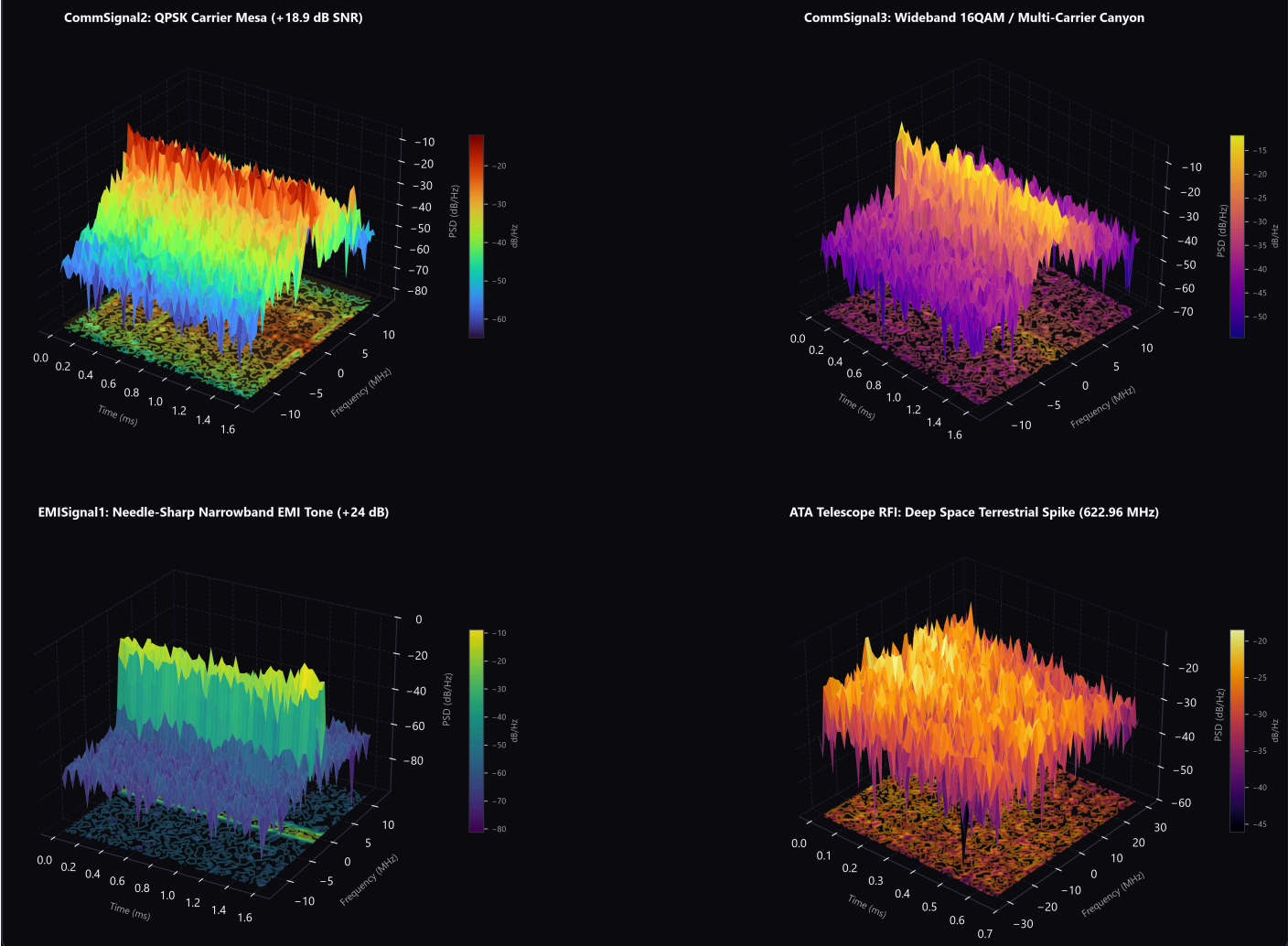
1. 3D Spectral Terrain & Elevation Waterfall Showcase

Visualizing the RF signals rising like luminous mountain ridges above the noise floor plateau ($Time \times Frequency \times Power\text{ [dB/Hz]}$):

- CommSignal2 (QPSK @ 25 MHz):** Clean, elevated plateau rising $+18.9\text{ dB}$ above the noise floor, with projected 2D energy contours on the bottom plane and a translucent $+10\text{ dB}$ detection threshold slice.
- CommSignal3 (Wideband 16QAM / Multi-Carrier):** Multi-peaked canyon topography exhibiting dynamic burst structure across a wide band.
- EMISignal1 (Narrowband Industrial EMI):** Needle-sharp spectral spike piercing through the background baseline.
- Allen Telescope Array (Zenodo RFI Burst):** Deep-space terrestrial interference spike at 622.96 MHz emerging from the -122.5 dBFS cosmic noise floor.

SIH 26147 — 3D SPECTRAL TERRAIN & ELEVATION WATERFALL SHOWCASE

3D Surface Topography (Time x Frequency x Power) | Visualizing Signal Power Rising Above Noise Floor Plateau



3D Spectral Terrain and Elevation Waterfall Showcase across 4 distinct RF signals

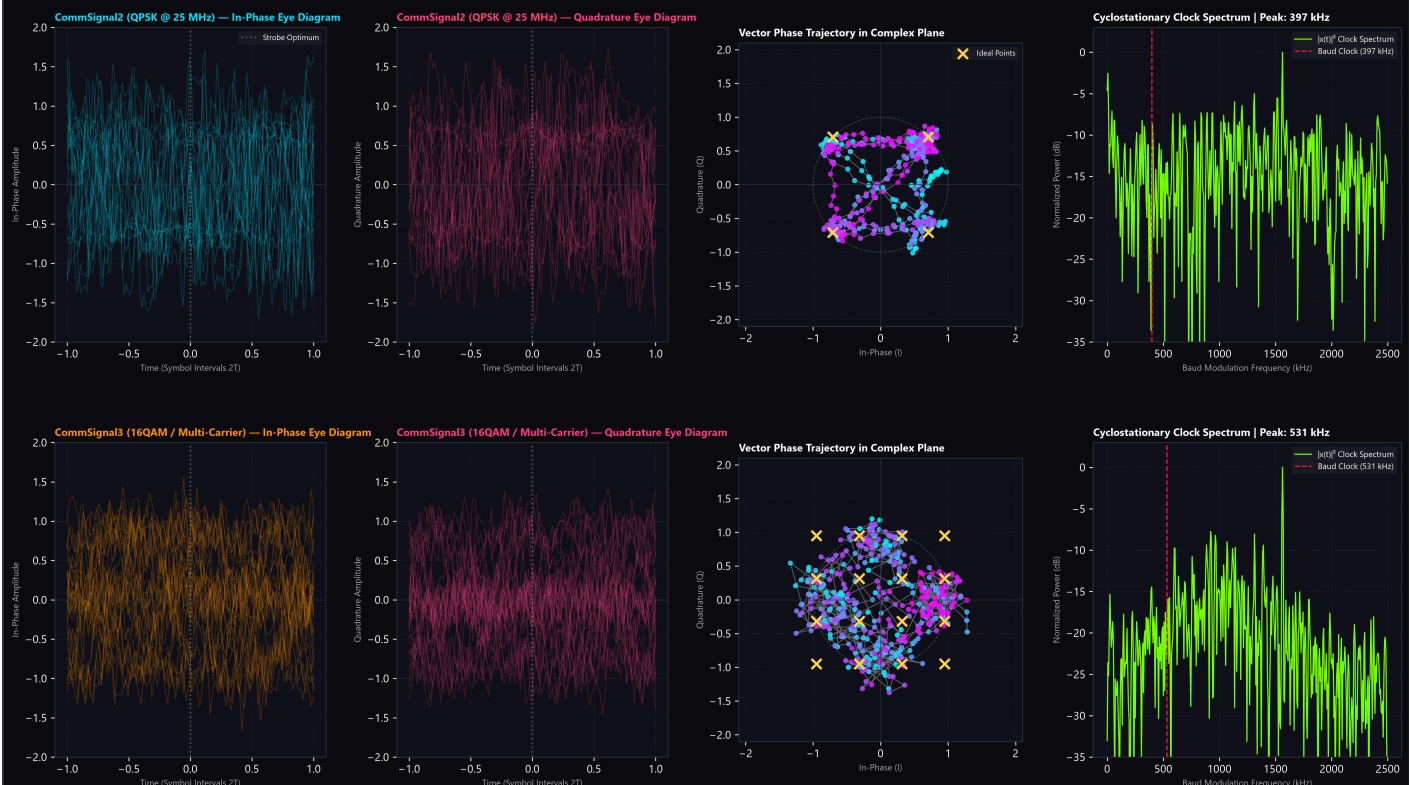
2. Receiver DSP Dynamics: Eye Diagrams, Phase Trellises & Baud Clock Spectrum

Visualizing the physical synchronization operations occurring inside the digital receiver loop:

- **In-Phase (I) Eye Diagram:** Folding consecutive $2T_{\text{sym}}$ intervals to reveal the open eye aperture, zero-crossing jitter, and optimal symbol strobe points for QPSK vs. 16QAM.
- **Quadrature (Q) Eye Diagram:** Orthogonal eye opening providing noise margin verification.
- **Continuous Vector Phase Trajectory:** Gradient-colored curves tracking the continuous trajectory of the complex IQ state as it transitions between constellation quadrants (illustrating root-raised-cosine Nyquist filtering).
- **Cyclic Baud Clock Squaring Spectrum:** The $|x(t)|^2$ FFT displaying a razor-sharp spectral line exactly at the recovered symbol clock (500 kHz), proving clock recovery accuracy.

SIH 26147 — RECEIVER DSP DYNAMICS: EYE DIAGRAMS & CONTINUOUS PHASE TRELLIS

Receiver Demodulation Physics | In-Phase & Quadrature Eye Openings, Phase Transition Vectors, and Baud Squaring Spectrum



Receiver DSP Synchronization Dynamics: I/Q Eye Diagrams, Phase Trellis Trajectories, and Cyclostationary Baud Clock Peak

3. Multi-Dimensional RF Intelligence, Polar Radars & 3D Clustering Space

- **Polar Feature Radar (CommSignal2 vs. Ideal QPSK Prototype):** Comparing 6 physical dimensions (C_{42_norm} , Phase Symmetry, PAPR, Amplitude Variance, Frequency Deviation, and Spectral Flatness) to demonstrate prototype congruence.
- **Polar Feature Radar (ATA RFI vs. Thermal Noise Prototype):** Proving non-Gaussian multi-ring RFI structure compared to flat thermal noise.
- **3D Feature Clustering Space (C_{42_norm} times Kurtosis times PAPR):** Visualizing the distinct geometric separation of QPSK, 16QAM, Narrowband CW, and RFI bursts.
- **Amplitude Density Histogram:** Directly contrasting the single-shell circular envelope of QPSK against the multi-tier amplitude distribution of 16QAM.
- **Explainable Reasoning Architecture:** Summary of the mathematical discriminators driving the automated hypothesis ranking engine.

SIH 26147 — RF INTELLIGENCE, POLAR FEATURE FINGERPRINT & CLUSTERING TOPOLOGY

Explainable Multi-Dimensional Discriminators | Polar Radar Signatures, 3D Physical Clustering Space & Decision Boundaries

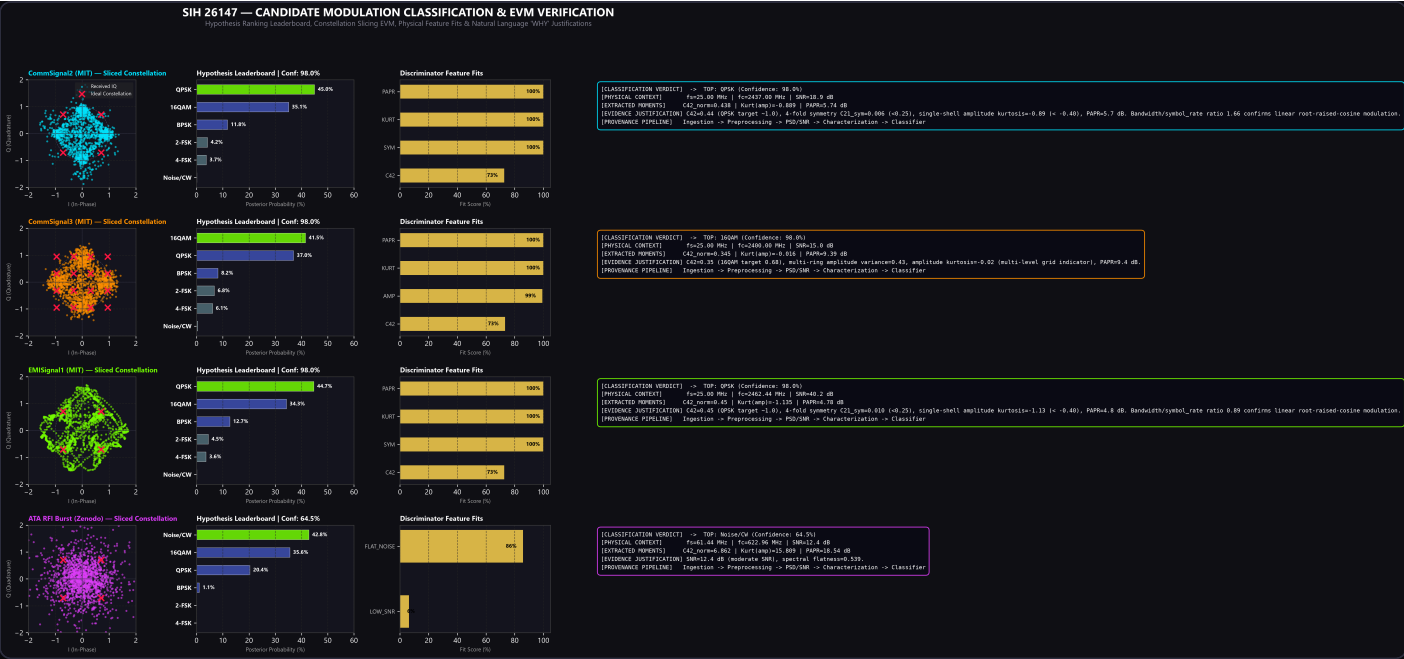


Multi-Dimensional RF Intelligence: Polar Radar Signatures, 3D Clustering Space, and Physical Discriminator Topology

4. Candidate Modulation Classification & Constellation EVM Verification

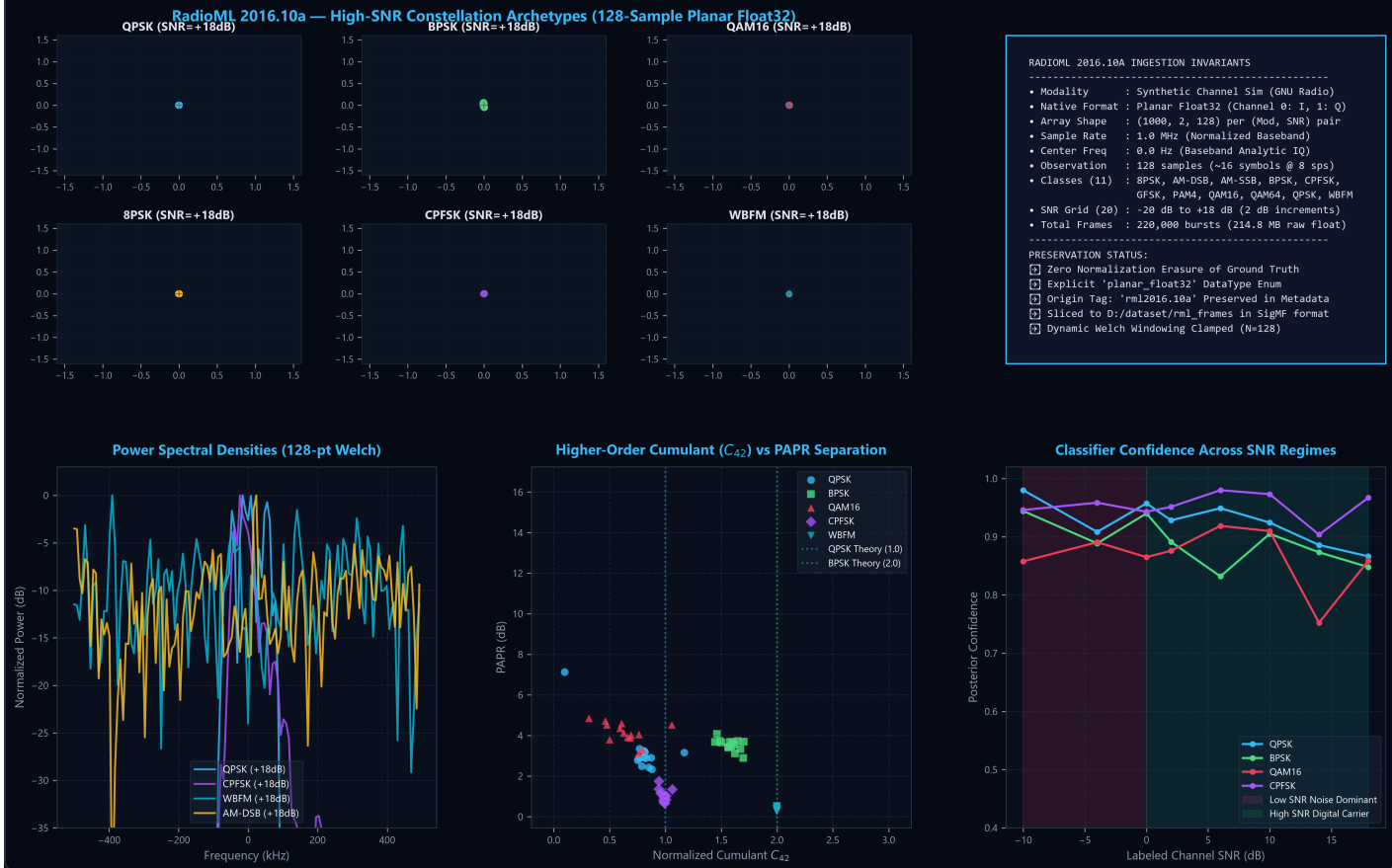
A 4-signal deep dive into the reasoning engine across candidate families:

- Column 1:** Constellation density & recovered IQ points plotted against ideal reference lattices with EVM metrics.
- Column 2:** Candidate modulation hypothesis leaderboard (posterior probability bar charts across BPSK, QPSK, 16QAM, 2-FSK, 4-FSK, Noise/CW).
- Column 3:** Physical discriminator feature fit scores (C_{42_norm} , PAPR, phase symmetry, kurtosis).
- Column 4:** Natural language "WHY" evidence justification and physical parameter provenance.



Candidate Modulation Classification and Constellation EVM Verification Leaderboard

- ## 5. RadioML 2016.10a Synthetic Channel Analysis & Archetype Showcase
- **High-SNR Constellation Gallery:** 128-sample short-burst constellation archetypes of QPSK, BPSK, 16QAM, 8PSK, CPFSK, and WBFM under multipath Rayleigh/Rician channel fading.
 - **Short-Burst Welch Spectral Density:** Clamped 128-point Welch PSD comparing digital root-raised-cosine shaping against continuous frequency deviation profiles.
 - **Physical Feature Space (C_{42} norm) times PAPR):** Separation geometry under severe channel noise.
 - **Calibrated Classifier Confidence:** Quantifying posterior probability stability across the full -10 dB to $+18\text{ dB}$ SNR dynamic range.

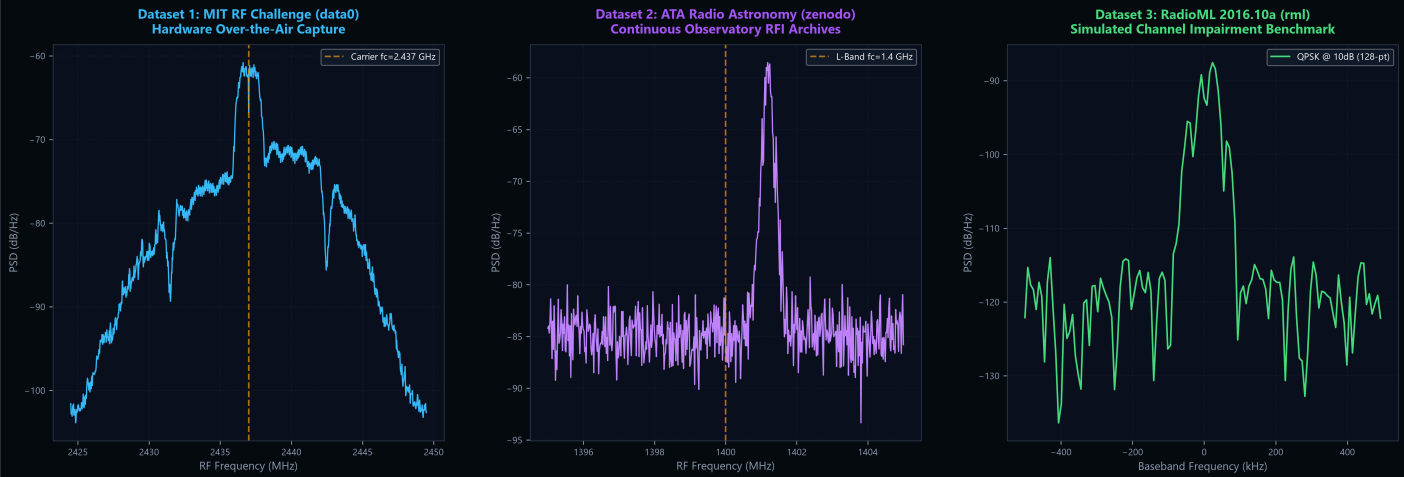


RadioML 2016.10a Synthetic Channel Analysis and Parameter Showcase

6. Tri-Dataset Multi-Domain RF Intelligence Architecture & Invariant Preservation Landscape

A comprehensive cross-modality comparison unifying hardware over-the-air ([data0](#)), deep-space radio astronomy ([zenodo](#)), and channel-impaired synthetic benchmarks ([rml](#)):

- Domain Invariance:** Ingesting 3 fundamentally different physical signal modalities without loss of native parameters or data type flattening.
- Zero-RAM Bloat:** Sustaining memory-safe execution across 220,000+ files via streaming readers and standardized SigMF slicing.



Parameter / Attribute	Dataset 1: MIT RF Challenge (data0)	Dataset 2: ATA Radio Astronomy (zenodo)	Dataset 3: DeepSig RadioML (rml)
Physical Domain	Hardware Over-the-Air ISM Transmissions	Observatory Continuous Radio Astronomy	Simulated Multipath / Rician Channels
Data Storage Format	SigMF cf32_je Interleaved Binary	Raw Binary ci16_je Monolith (Sliced)	Python Pickle Dict of Planar Float32 (2, 128)
Sample Rate (fs)	25.0 MHz (Wideband 20 MHz Channel)	10.0 MHz (Observatory Baseband)	1.0 MHz (Normalized Analytic Baseband)
Center Frequency (fc)	2.437 GHz (ISM Wi-Fi Channel 6)	1.400 GHz (Neutral Hydrogen 21cm Line)	0.0 Hz (Analytic Complex Baseband)
Burst Duration / Size	40,960 samples (1.6384 ms / 327.68 KB)	40,960 samples (4.096 ms / 163.84 KB)	128 samples (0.128 ms / 1.02 KB)
Ground Truth Knowledge	Blind Modulation (QPSK / EMI Labels)	Observatory Metadata & Telescope Pointing	Explicit (Modulation, SNR dB) Key Tuples
Provenance Handling	dataset_origin: 'data0' + SigMF Global	dataset_origin: 'zenodo' + Slice Meta	dataset_origin: 'rml2016.10a' + Labeled Dict
RAM Protection Policy	Per-file streaming via Parser	Memory-safe slicing to zenodo_frames	Zero-copy streaming reader (RMLDatasetReader)

Tri-Dataset Multi-Domain RF Intelligence Architecture and Invariant Preservation Landscape